

VERITAS AI — Admissions Essay Forensics Engine

Scientific Multi-Signal Text Verification, Sentence-Level Interpretability & Bias-Aware Fairness Architecture

98.40%	0.9995	3.30%	96.70%	0.0133	0.00%
TEST ACCURACY	ROC-AUC SCORE	ESL FALSE POSITIVE	ESL SPECIFICITY	BRIER LOSS	LLM DEPENDENCY

1. Executive Overview & The Non-LLM Verdict Rule

Modern commercial AI detectors rely on brittle perplexity thresholds or recursive LLM querying that produce unacceptable false positives against international applicants (ESL) while offering zero forensic explainability. **Veritas AI solves this through a non-LLM, multi-signal mathematical architecture.** Authenticity verdicts are determined strictly by deterministic statistical extractors, semantic trajectory geometry, and a fine-tuned transformer attention model. Large Language Models (Groq LLaMA 3.3 70B) are restricted entirely to post-hoc translation—narrating mathematical evidence into plain English for admissions officers without ever voting on or altering authenticity scores.

2. The 4 Orthogonal Analytical Signals

Veritas extracts 8 mathematical dimensions across 4 complementary linguistic signals for every sentence:

SIGNAL	METHODOLOGY & MATHEMATICAL FORMULATION	FORENSIC WEIGHT (β)
Signal A Vocabulary Signature	Monroe et al. Dirichlet-prior log-odds ratio calculation comparing essay vocabulary against authentic human Common App baseline datasets. Identifies hallmark synthetic n-grams (e.g. 'testament to', 'pivotal role', 'delve into', 'fostering') with exact Z-score deviations.	$\beta = +1.54$ High AI Skew
Signal B Narrative Trajectory	Sentence-by-sentence dense embedding via MiniLM-L6-v2. Computes cosine similarity velocity across consecutive sentence pairs. Human storytelling exhibits non-linear narrative leaps and conversational tangents (high variance), while LLM outputs demonstrate monotonic, uniform semantic pacing.	$\beta = +0.57$ Uniformity Penalty
Signal C Stylometrics & Burstiness	Quantifies structural burstiness and syntactic complexity: sentence length variance, Type-Token Ratio (TTR), Flesch-Kincaid reading grade, formulaic discourse transition counts ('Moreover', 'In conclusion'), and passive voice construction indicators.	$\beta = -0.48$ Burstiness Protector
Signal D Supervised Classifier	Fine-tuned DeBERTa-v3-base sequence classifier trained on curated human admissions essays and frontier LLM variants (ChatGPT, Claude, Gemini, LLaMA). Operates with cross-attention token modeling to detect subtle statistical token transitions at the sentence level.	$\beta = +6.39$ Primary Transformer

Interactive Forensic Heatmap: Unlike traditional monolithic detectors that provide only a single document percentage, Veritas AI scores every sentence independently, color-coding sentences into 3 risk bands: **HIGH_AI** ($\geq 70\%$), **UNCERTAIN** (40–70%), and **HUMAN** ($< 40\%$). Admissions officers can click any sentence to audit its 4-signal telemetry breakdown.

3. Calibrated Combiner & Decision Boundary

All extracted features are integrated using a trained **multivariate logistic regression combiner** with calibrated intercept ($\beta_0 = -3.92$). The model maps 8-dimensional feature vectors into an exact posterior probability $P(AI | x)$. When evaluating essays, Signal C's length burstiness parameter ($\beta = -0.48$) directly prevents false positives on expressive human writing, while Signal D and Signal A identify synthetic homogeneity.

4. End-to-End Technology Stack

LAYER	TECHNOLOGY / FRAMEWORK	ARCHITECTURAL ROLE & FUNCTION
Core Backend	FastAPI (Python 3.11) + Uvicorn	High-throughput asynchronous ASGI microservice orchestration with instant endpoint response and lazy singleton loading.
Deep Learning	PyTorch + Hugging Face Transformers	DeBERTa-v3 sequence classification and sentence-level token attention scoring with CUDA acceleration / optimized CPU fallback.
NLP & Semantics	Sentence-Transformers + spaCy + textstat	MiniLM-L6-v2 cosine trajectory modeling, robust sentencizer tokenization, Flesch reading ease, and lexical TTR analytics.
Combiner Engine	Scikit-Learn + NumPy + Joblib	L2-regularized multivariate logistic calibration translating multi-signal telemetry into calibrated posterior probabilities.
Explainability	Groq SDK (LLaMA 3.3 70B Versatile)	Fast post-hoc natural language synthesis translating mathematical indicators into transparent, plain-English audit summaries.
Frontend UI/UX	React 18 + Vite + Tailwind CSS + Lucide	Professional DAW-inspired forensic workbench with interactive heatmap, radar telemetry, and dual-mode high-contrast themes.

5. What Makes Veritas AI Truly Unique (Competitive Moat)

1. Absolute Non-LLM Verdict Integrity	Unlike GPT-wrapper detectors that hallucinate verdicts, Veritas uses zero LLM inference to compute authenticity. The evaluation is 100% reproducible and grounded in transparent statistical and geometric evidence.
2. Rigorous Non-Native (ESL) Applicant Equity & Protection	Standard commercial tools penalize international students (30%+ false positive rates) because ESL writing uses structured transition markers ('Moreover', 'In conclusion'). Veritas achieves a benchmarked 96.7% Specificity (3.3% FPR) by compensating with narrative leap analysis.
3. Granular Sentence-Level Heatmap with DAW Visual Telemetry	Admissions officers receive visual evidence mapped across Introduction, Body, and Conclusion paragraphs, rather than an unhelpful single percentage.
4. Post-Hoc Groq Translation Without Black-Box Bias	Admissions committees can audit exactly why a sentence was flagged with transparent plain-English telemetry explanations generated in <500ms.

Conclusion: Veritas AI establishes a new gold standard for academic and admissions integrity—delivering state-of-the-art 98.4% detection accuracy, provable fairness for international applicants, and complete scientific interpretability.